

BUNBURY, AustraliaWMO: **946040**Lat: **33.3567S**Lon: **115.6447E**Elev: **20**StdP: **14.69**Time Zone: **8.00 (E08)**Period: **99-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.6	39.6	34.7	29.5	44.4	36.7	31.9	45.8	25.8	58.7	23.4	58.8	1.4	90	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.5	92.8	67.8	89.6	67.1	86.7	66.6	71.5	84.2	70.0	82.5	68.7	81.3	9.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.1	103.2	75.6	66.2	96.5	74.3	64.5	90.9	73.5	35.3	84.7	34.0	82.2	32.9	81.2	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	18.7	16.8	DB	33.5	99.5	1.9	2.6	32.1	101.3	31.0	102.9	29.9	104.3	28.5	106.2	(4)
(5)			WB	32.7	73.9	1.9	2.0	31.4	75.4	30.3	76.6	29.2	77.7	27.9	79.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	62.2	72.0	72.2	69.1	63.8	58.8	55.5	53.9	54.2	55.6	59.0	64.7	68.2	(6)
(7)		DBStd	8.36	5.78	5.74	5.95	5.08	4.40	4.11	4.16	4.04	4.18	4.82	5.65	6.29	(7)
(8)		HDD50	30	0	0	0	0	1	4	12	8	5	0	0	0	(8)
(9)		HDD65	1876	9	8	28	81	197	286	346	334	283	195	74	35	(9)
(10)		CDD50	4477	680	622	592	414	273	168	131	139	172	280	441	565	(10)
(11)		CDD65	849	224	210	155	45	5	0	0	0	0	10	65	135	(11)
(12)		CDH74	8088	2312	2048	1336	278	23	0	0	0	9	91	588	1403	(12)
(13)		CDH80	2947	931	798	442	47	1	0	0	0	1	14	173	540	(13)
(14)	Wind	WSAvg	8.3	10.0	9.7	8.7	6.9	6.4	6.9	7.6	7.4	8.3	8.4	9.4	9.5	(14)
(15)	Precipitation	PrecAvg	29.4	0.4	0.3	0.9	1.5	4.3	5.2	5.7	4.6	3.4	1.3	1.1	0.6	(15)
(16)		PrecMax	38.4	2.4	1.9	3.2	4.0	8.8	9.1	10.4	6.3	7.0	3.1	2.8	4.2	(16)
(17)		PrecMin	18.8	0.0	0.0	0.0	0.0	0.8	1.3	2.2	1.1	0.9	0.2	0.1	0.0	(17)
(18)		PrecStd	4.8	0.6	0.4	0.9	1.1	2.0	2.2	1.9	1.2	1.5	0.7	0.7	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.5	95.7	94.1	85.1	77.0	70.2	67.2	69.6	74.0	81.9	90.2	95.0	(19)
(20)			MCWB	68.2	69.4	68.5	64.9	61.0	58.4	58.8	59.1	59.2	64.0	65.3	67.0	(20)
(21)		2%	DB	92.3	92.0	88.7	80.7	72.9	67.6	65.1	66.2	68.7	76.2	85.0	90.1	(21)
(22)			MCWB	67.9	68.4	66.8	63.9	60.5	59.1	57.7	59.0	59.0	62.2	64.1	66.7	(22)
(23)		5%	DB	88.7	88.2	84.8	77.1	70.4	65.8	63.6	64.3	65.9	71.9	80.6	86.0	(23)
(24)			MCWB	67.4	67.1	65.7	63.4	60.6	58.6	57.1	57.9	58.3	61.0	63.6	65.6	(24)
(25)		10%	DB	85.1	85.1	81.2	74.0	68.3	64.3	62.0	62.6	64.0	68.7	76.6	82.1	(25)
(26)			MCWB	66.9	66.7	65.0	63.0	60.2	57.8	56.1	56.7	57.4	59.9	62.4	64.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.6	74.3	71.7	69.6	66.6	65.6	62.1	63.2	63.0	66.5	68.4	71.6	(27)
(28)			MCD B	84.8	86.8	83.3	75.4	70.7	66.4	65.0	65.8	68.5	76.5	80.0	84.3	(28)
(29)		2%	WB	71.5	71.4	69.9	67.7	64.6	62.1	59.8	60.8	61.1	64.4	66.9	69.3	(29)
(30)			MCD B	84.7	84.6	80.5	73.9	68.6	65.0	63.2	64.2	66.1	72.2	78.2	84.0	(30)
(31)		5%	WB	69.7	69.8	68.5	66.0	62.9	60.3	58.2	59.0	59.4	62.5	65.4	67.5	(31)
(32)			MCD B	83.1	83.1	79.5	72.7	67.4	63.9	62.1	63.0	64.5	69.5	76.3	81.8	(32)
(33)		10%	WB	68.0	68.2	67.1	64.4	61.3	58.6	56.7	57.3	58.0	60.8	63.8	65.8	(33)
(34)			MCD B	82.8	81.6	78.2	72.4	66.6	62.8	60.9	61.5	63.2	67.8	74.4	79.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	23.0	22.5	21.4	19.6	18.9	17.1	15.8	16.2	16.5	18.2	19.9	21.5	(35)
(36)			MCDBR	27.9	26.4	25.6	23.2	21.8	19.0	16.7	18.0	19.2	23.2	25.8	27.5	(36)
(37)			MCWBR	10.9	9.9	10.3	11.0	12.5	12.3	11.6	12.0	12.2	12.8	11.6	11.1	(37)
(38)		5% WB	MCDBR	23.0	22.1	20.2	17.2	16.2	14.4	13.3	14.3	16.1	18.9	21.2	24.1	(38)
(39)			MCWBR	11.2	10.1	10.1	10.6	11.9	11.5	10.8	11.4	11.7	12.1	11.5	11.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.332	0.334	0.321	0.314	0.309	0.308	0.304	0.313	0.328	0.325	0.331	0.327	(40)	
(41)		taud	2.588	2.588	2.626	2.640	2.625	2.604	2.614	2.576	2.513	2.538	2.546	2.577	(41)	
(42)		Ebn at Noon	319	312	305	291	275	266	274	286	298	311	318	322	(42)	
(43)		Edh at Noon	33	32	29	25	23	22	23	27	32	33	34	34	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2524	2251	1815	1290	936	755	806	1066	1432	1898	2294	2536	(44)	
(45)		RadStd	130	89	89	80	65	47	45	63	83	93	134	126	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)	N/A	N/A	N/A	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page